

DW_ram_2r_2w_s_dff

Sync. Write, Async. Read, 4-port (2rd/2wr) RAM (FF-Based)

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameter controlled data width
- Parameter controlled address width (controls memory size)
- Synchronous static memory
- Parameter controlled reset mode (synchronous or asynchronous)

Revision History

Description

DW_ram_2r_2w_s_dff implements a 4-port synchronous write, asynchronous read flip-flop based RAM with two write ports and two read ports. If both write ports attempt to write to the same RAM address at the same time, write port 1 is written and the data from write port 2 is ignored.

The inputs `en_w1_n` and `en_w2_n` are used to control when data is to be written to the RAM array. When `en_w1_n` is low (logic zero) at the rising edge of `clk`, the data on `data_w1` is written to RAM location `addr_w1`. When `en_w2_n` is low (logic zero) at the rising edge of `clk` and `addr_w2` not equal to `addr_w1`, the data on `data_w2` is written to RAM location `addr_w2`. This implies that write port 1 has priority over write port 2 when both write port attempt to write to the same address in the same clock cycle (if `addr_w1` = `addr_w2` and both `en_w1_n` and `en_w2_n` are low, the memory location will be written with `data_w1` while `data_w2` will be ignored)

The inputs `en_r1_n` and `en_r2_n` are used to enable the read ports 1 and 2. When `en_r1_n` is inactive (logic one), `data_r1` is driven to all zeros. When `en_r1_n` is active (logic zero) `data_r1` is selected to contain the data in location `addr_r1` of the RAM. When `en_r2_n` is inactive (logic one), `data_r2` is driven to all zeros. When `en_r2_n` is active (logic zero) `data_r2` is selected to contain the data in location `addr_r2` of the RAM.

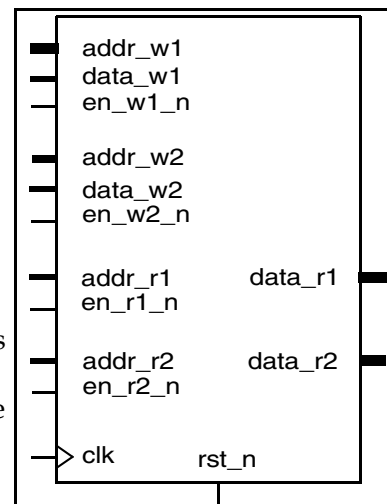


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
clk	1 bit	Input	Clock
rst_n	1 bit	Input	Reset, active low
en_w1_n	1 bit	Input	Write port 1 enable, active low
addr_w1	<i>addr_width</i>	Input	Write port 1 address
data_w1	<i>width</i>	Input	Write port 1 data in
en_w2_n	1 bit	Input	Write port 2 enable, active low

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
addr_w2	<i>addr_width</i>	Input	Write port 2 address
data_w2	<i>width</i>	Input	Write port 2 data in
en_r1_n	1 bit	Input	Read port 1 enable, active low
addr_r1	<i>addr_width</i>	Input	Read port 1 address
data_r1	<i>width</i>	Output	Read port 1 data out
en_r2_n	1 bit	Input	Read port 2 enable, active low
addr_r2	<i>addr_width</i>	Input	Read port 2 address
data_r2	<i>width</i>	Output	Read port 2 data out

Table 1-2 Parameter Description

Parameter	Values	Description
width	1 to 8192 Default: 8	Data width
addr_width	1 to 12 Default: 3	Address bus width - which controls memory depth.
rst_mode	0 or 1 Default: 0	Determines the reset methodology: 0: <i>rst_n</i> asynchronously initializes the RAM 1: <i>rst_n</i> synchronously initializes the RAM

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	DesignWare

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:
``define DW_SUPPRESS_WARN`
- Or, include a command line option to the simulator, such as:
`+define+DW_SUPPRESS_WARN` (which is used for the Synopsys VCS simulator)

The warning messages for this model include the following:

- If values other than 1 or 0 are present on a clock port, the following message is displayed:

```
WARNING: <instance_path>.<clock_name>_monitor:  
        at time = <timestamp>, Detected unknown value, x, on <clock_name> input.
```

To suppress only this warning message for all DWBB components, use the following macro:

- Define the DW_DISABLE_CLK_MONITOR macro. You can define this macro in the following ways:
 - Specify the Verilog preprocessing macro in Verilog code:
``define DW_DISABLE_CLK_MONITOR`
 - Or, include a command line option to the simulator, such as:
`+define+DW_DISABLE_CLK_MONITOR` (which is used for the Synopsys VCS simulator)

This message is also suppressed using the DW_SUPPRESS_WARN macro explained earlier.

Related Topics

- [Memory – Synchronous RAMs Listing](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,WORK,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;

entity DW_ram_2r_2w_s_dff_inst is
    generic (
        width : INTEGER := 8;
        addr_width : INTEGER := 3;
        rst_mode : INTEGER := 0
    );
    port (
        clk : in std_logic;
        rst_n : in std_logic;
        en_w1_n : in std_logic;
        addr_w1 : in std_logic_vector(addr_width-1 downto 0);
        data_w1 : in std_logic_vector(width-1 downto 0);
        en_w2_n : in std_logic;
        addr_w2 : in std_logic_vector(addr_width-1 downto 0);
        data_w2 : in std_logic_vector(width-1 downto 0);
        en_r1_n : in std_logic;
        addr_r1 : in std_logic_vector(addr_width-1 downto 0);
        data_r1 : out std_logic_vector(width-1 downto 0);
        en_r2_n : in std_logic;
        addr_r2 : in std_logic_vector(addr_width-1 downto 0);
        data_r2 : out std_logic_vector(width-1 downto 0)
    );
end DW_ram_2r_2w_s_dff_inst;
```

architecture inst of DW_ram_2r_2w_s_dff_inst is

```
    component DW_ram_2r_2w_s_dff
        generic (
            width : INTEGER := 8;
            addr_width : INTEGER := 3;
            rst_mode : INTEGER := 0
        );
        port (
            clk : in std_logic;
            rst_n : in std_logic;

            en_w1_n : in std_logic;
            addr_w1 : in std_logic_vector(addr_width-1 downto 0);
            data_w1 : in std_logic_vector(width-1 downto 0);

            en_w2_n : in std_logic;
            addr_w2 : in std_logic_vector(addr_width-1 downto 0);
```

```
    data_w2 : in std_logic_vector(width-1 downto 0);

    en_r1_n : in std_logic;
    addr_r1 : in std_logic_vector(addr_width-1 downto 0);
    data_r1 : out std_logic_vector(width-1 downto 0);

    en_r2_n : in std_logic;
    addr_r2 : in std_logic_vector(addr_width-1 downto 0);
    data_r2 : out std_logic_vector(width-1 downto 0)
  );
end component;

begin

  -- Instance of DW_ram_2r_2w_s_dff
  U1 : DW_ram_2r_2w_s_dff
    generic map ( width => width,
                  addr_width => addr_width,
                  rst_mode => rst_mode )

    port map ( clk => clk, rst_n => rst_n,
              en_w1_n => en_w1_n, addr_w1 => addr_w1, data_w1 => data_w1,
              en_w2_n => en_w2_n, addr_w2 => addr_w2, data_w2 => data_w2,
              en_r1_n => en_r1_n, addr_r1 => addr_r1, data_r1 => data_r1,
              en_r2_n => en_r2_n, addr_r2 => addr_r2, data_r2 => data_r2 );

end inst;
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_ram_2r_2w_s_dff_inst(
    clk, rst_n,
    en_w1_n, addr_w1, data_w1,
    en_w2_n, addr_w2, data_w2,
    en_r1_n, addr_r1, data_r1,
    en_r2_n, addr_r2, data_r2 );

parameter width = 8;
parameter addr_width = 3;
parameter rst_mode = 0;

input clk;
input rst_n;
input en_w1_n;
input [addr_width-1 : 0] addr_w1;
input [width-1 : 0] data_w1;
input en_w2_n;
input [addr_width-1 : 0] addr_w2;
input [width-1 : 0] data_w2;
input en_r1_n;
input [addr_width-1 : 0] addr_r1;
output [width-1 : 0] data_r1;
input en_r2_n;
input [addr_width-1 : 0] addr_r2;
output [width-1 : 0] data_r2;

// Instance of DW_ram_2r_2w_s_dff
DW_ram_2r_2w_s_dff #(width, addr_width, rst_mode)
U1 (
    .clk(clk), .rst_n(rst_n),

    .en_w1_n(en_w1_n), .addr_w1(addr_w1), .data_w1(data_w1),
    .en_w2_n(en_w2_n), .addr_w2(addr_w2), .data_w2(data_w2),

    .en_r1_n(en_r1_n), .addr_r1(addr_r1), .data_r1(data_r1),
    .en_r2_n(en_r2_n), .addr_r2(addr_r2), .data_r2(data_r2)
);

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet that shows the most recent changes, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
March 2024	DWBB_202312.2	■ Restored correct maximum values for <i>adder_width</i> in Table 1-2 on page 2
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Adjusted content and title of “ Suppressing Warning Messages During Verilog Simulation ” on page 3 and added the DW_SUPPRESS_WARN macro
October 2019	DWBB_201903.5	■ Added the “Disabling Clock Monitor Messages” section ■ Added this Revision History table and the document links on this page

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